15 **(a)** On the axes in the Printed Answer Booklet, sketch the graph of $y = 3^x$. **[2]**

(b) Determine the **exact** coordinates of the points of intersection of the curve $y = 3^x$ with the curve $y = 3^{2x} - 6 \times 3^x + 12$. **[5]**

Turn over for question 16

